

Electric Vehicle Market Analysis

Key insights

- BEVs with 197,146 vehicles ($\approx 80\%$) dominate the market compared to Plug-in Hybrid Electric Vehicles (PHEV) 50,198 ($\approx 20\%$) indicating a strong shift toward fully electric technology over hybrid models.
- EV adoption increased sharply from 2016 onward, peaking around 2023 ($\approx 59\text{K}$ vehicles). A small decline in 2024–2025 could suggest data lag or market stabilization after rapid growth.
- The highest concentration of EVs is seen in a few key states— Washington leads significantly with $\sim 246\text{K}$ EVs, followed by scattered adoption across other states. EV growth is mainly strong in states with better support and charging facilities.
- Tesla is the clear market leader, holding over half of the total EV market, indicating brand dominance and consumer trust in Tesla's technology.
- premium electric vehicles, despite their higher prices, generally offer a lower or moderate electric range compared to mid and low-priced models. This suggests that higher cost does not always mean better range performance. Premium models may focus more on luxury features, design, or performance rather than maximizing range efficiency. Overall, the relationship indicates that electric range depends more on vehicle type and design priorities than on price category.
- Battery Electric Vehicles (BEVs) offer higher average range ($\approx 48\text{ km}$) than Plug-in Hybrid Electric Vehicles (PHEVs) ($\approx 31\text{ km}$).